

Raayan

Moury
1, 2, 5, 7.

Health
1st test

6 marks

Answer any one of the following questions:

1. (a) The Practitioner provided medical care has a small role in the historical decline of population

Mortality rate-Explain the above statement with practical examples.

(b) Explain the two different theories of the schooling.

2. (a) How would you derive the demand curve for visit?

(b) Explain the determinants of health care demand.

124578

Define production function. Describe the production function of health and marginal production function of health care. Would the production function of health eventually bend downward?

Production function:

Production function refers to the functional relationship between the quantity of good produced (output) and the factors of production (input)

$$Q = f(L, C)$$

where, Q = Quantity of output

C = Capital

L = Labor

Hence, the level of output (Q) depends on the quantities of different inputs (L, C) available to them.

Production function of Health:

Production function of health refers to the functional relationship between health care inputs and health.

$$HS = \text{Health status (HS)} = f(\text{Health care, Lifestyle, Environment, Human biology})$$

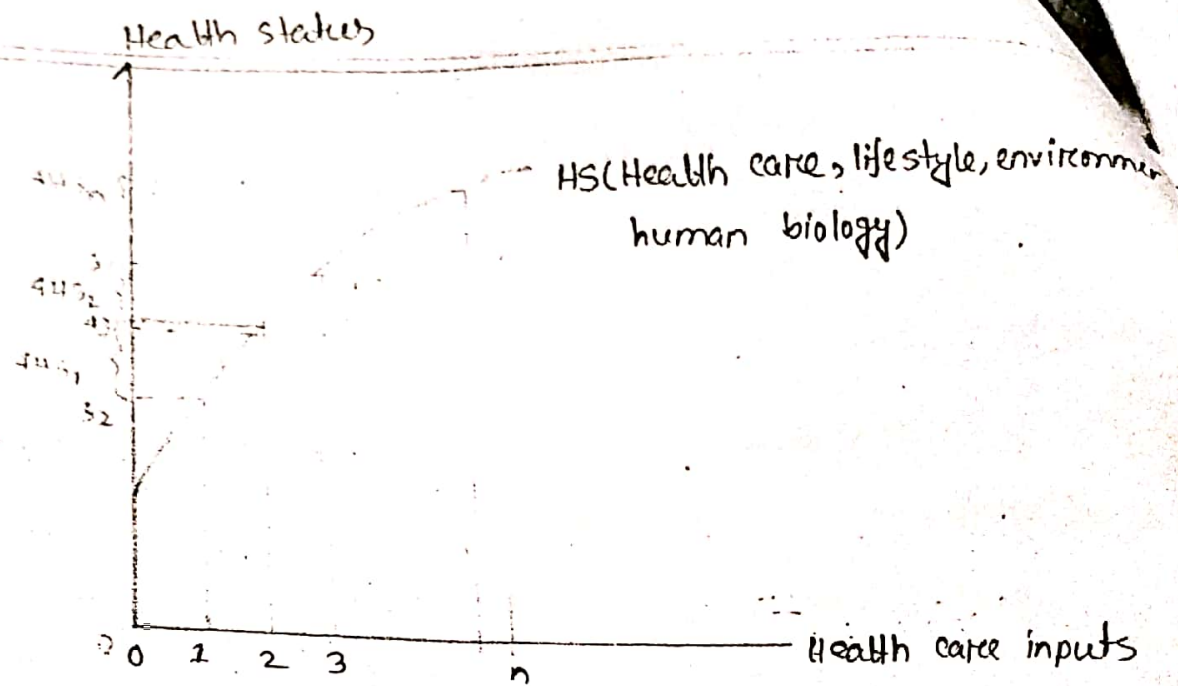
The marginal contribution of health care is its marginal product, meaning the increment to health caused by one

extra unit of Health care, holding all other inputs constant

The distinction between total health production function and marginal health production function is illustrated in the following figure. Figure A exhibits a theoretical health status production function for the population. Health status is shown here to be an increasing function of health care.

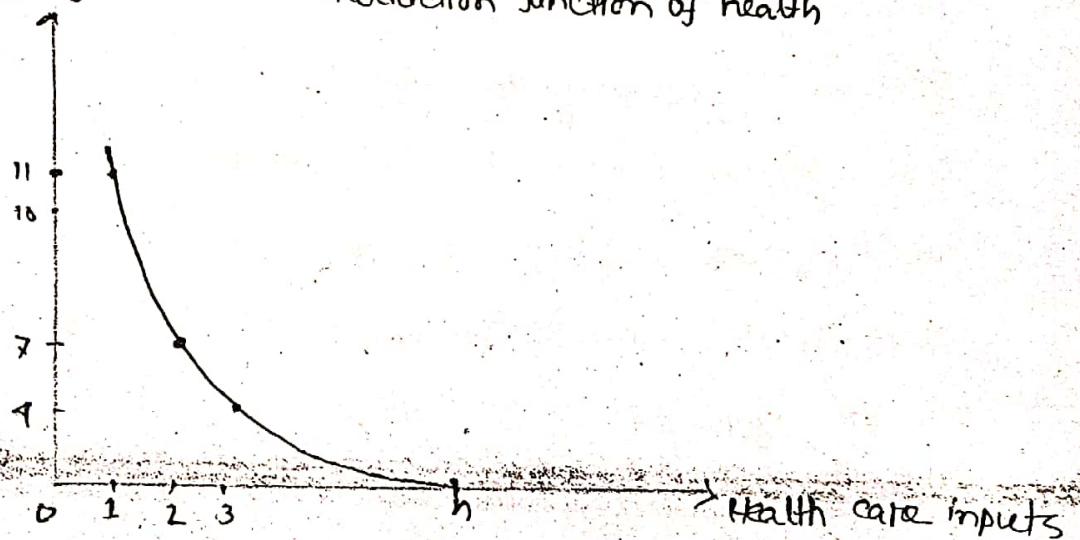
A production function describes the relationship of flows of inputs and flows of outputs over a specified time period. To simplify the depiction, we have reduced all health care inputs into one scale called Health care. In reality, health care consists of many different health care inputs. Conceptually, the health care measure, HC may be thought of as an aggregate of all these types of health care, the aggregation being based on dollar values.

Increasing health care from zero to one unit in Figure A improves health status by ΔHS_1 , the first unit's marginal product. Numerically, the first unit of health care has increased the health status index from 32 to 43 $\Delta HS_1 = 11$ Health status units. The next unit of medical care delivers a marginal product of $\Delta HS_2 = 7$ and so on.



marginal product of health care

4. Production function of health



B. Marginal product of health care

These marginal products are diminishing in size, illustrating the law of diminishing marginal returns. If society employs a total of n units of health care, then

the total contribution of health care is the sum of the marginal products of each of the n units. This total contribution as shown AB may be substantial. However, the marginal product of the n th unit of medical care is ΔHS_n and it is small. In fact, we are nearly on the "flat of the curve." Marginal product is graphed on figure B.

The health production function eventually bend downward. This is a logical possibility under at least two scenarios.

(i) multiple drugs may have adverse interactions.

(ii) Iatrogenic disease means provider caused disease.

This is an inevitable by product of many ~~medical~~ ^{care/inevitable} interventions. For example, each surgery has its risks.

Medical scientists such as Cochrane (1972), have pressed the case that much medical care as often practiced has only weak scientific basis, making iatrogenesis a real probability.

Ind. Historical

Practitioners provided medical interventions played a small perhaps negligible role in the historical decline on population mortality rates. Explain this statement with the help of empirical evidence.

Many presume that the declines in the mortality rates were due to improvements in medical science provided to the public through medical practice but counter-arguments to this proposition bring it into question. The argument can be illustrated for the case of respiratory tuberculosis and a group of these three upper respiratory diseases - bronchitis, pneumonia and influenza. Mortality for these diseases fell to relatively low levels prior to availability of effective medical interventions, whose availability occurred respectively after 1930 and for some cases well into the 1950s and 1960s. This picture is shared by water borne diseases. About 95 percent of the mortality due to cholera, diarrhoea and dysentery occurred prior to 1930s when intravenous therapies became available. Typhoid and typhus mortality already had fallen to low levels by the beginning of the twentieth century. In most cases, as is shown, the availability of the effective medical

declines in deaths from "pneumonia - diarrhea" complex.
Fuchs concluded:

It is important to realize that germs can play
almost no role in this decline unless we know the
precise causes. It is believed that rising living
standards, the spread of literacy and education and a
substantial fall in the birth rate all played a part."

In the 1930s, antimicrobial drugs were introduced.

Between 1935 and 1950 the fall in infant death rates
accelerated. Fuchs proposed that during this period "both
medical advances and rising living standards contributed
to the reduction in infant deaths".

~~2nd section~~
L Explain two different theories of schooling.

First, Michael Grossman's theory of demand entails a central role for education. Grossman contends that better educated person tends to be economically more efficient producers of health status. Better educated people understand the technology or have the know-how needed to stay healthy. They also know better how to use medical and other market inputs and their own time to ~~perfectly~~ produce health. Under this view, the marginal transfer of resources makes good sense.

Much of the available research concentrates formal schooling. Other health economists have put forward hypotheses under which schooling and health status are correlated only because they are both related to one or more other factors.

Second, Victor Fuchs has suggested a theory that involves certain microeconomic aspects of time discounting that is comparing the future to the present.

Decision makers must discount future benefits and costs of a social investment to make them comparable in terms of present values. A decision maker with a high discount rate will tend to prefer projects with immediate payoffs versus long term projects.

What is health economics? Describe the relevance of health economics/why do we study health economics?

Health economics:

Health economics is a branch of economics concerned with issues related to efficiency, effectiveness, value and behaviour in the production and consumption of health and health care. Health economics studies how resources are allocated to and within the health economy. The production of health care and its distribution across populations fall within this definition. Kenneth Arrow is the father of health economics.

The relevance of health economics:

The study of health economics is important and interesting in three related ways:

- (i) the size of the contribution of health sector to the overall economy;
- (ii) the national policy concerns resulting from the importance many people attach to the economic problems they face in pursuing and maintaining their health; and
- (iii) the many health issues that have a substantial economic element.

element.

Features of economic analysis

Many distinctive features of economics might be identified but we emphasize four.

- (i) Scarcity of social resources
- (ii) Assumption of rational decision making
- (iii) Concept of marginal analysis
- (iv) Use of economic models.

(i) Scarcity of resources:

Economic analysis is based on the premise that individuals must give up some of one resource in order to get some of another. At the national level, this means that increasing shares of GDP going to health care ultimately imply decreasing shares going for other uses. Economists view time as the ultimate scarce resource. Individuals sell their time for wages, and many individuals will refuse overtime work even if offered more than their normal

wage rate:

(ii) Rational decision making:

Economists typically approach problems of human economic behaviour by assuming that the decision maker is a rational being. Rationality is effectively defined as "making choices that best further one's own ends given one's resource constraints." Some behaviours may appear irrational.

(iii) Marginal analysis:

Mainstream economic analyses feature reasoning at the margin. To make an appropriate choice, decision makers must understand the cost as well as the benefit of the next or marginal unit.

(iv) Use of models:

Finally economics characteristically develops models to depict its subject matter. The models may be described in words, graphs or mathematics.

✓ Does economics apply to health and health care?
Does price matter?

If economics studies how scarce resources are used to produce goods and services and then how these goods and services are distributed, then clearly economics applies. Certainly health care resources are scarce; in fact, it is their cost that concerns most people. There is no ~~reason~~ question that health care is produced and distributed.

However, much of health care simply does not fit this emergency image. A considerable amount of health care is elective, meaning that patients have some choice over whether and when to have the diagnostics or treatment. Much health care is even routine, involving problems such as colds, respiratory infections, back pain and diagnostics. The patients often have prior experience with the concerns. Furthermore, even in a real emergency, the consumer has agents to make or help make decisions.

on his or her behalf.

Does price matter?

Many have argued that health care is so different from other goods that consumers do not respond to financial incentives. Following figure examines the use of ambulatory mental health and medical care where amounts of health care consumed are measured along the horizontal axis. These amounts are scaled in percentage terms from zero to 100 percent, where 100 percent reflects the average level of care consumed by the group that used the most care on average.

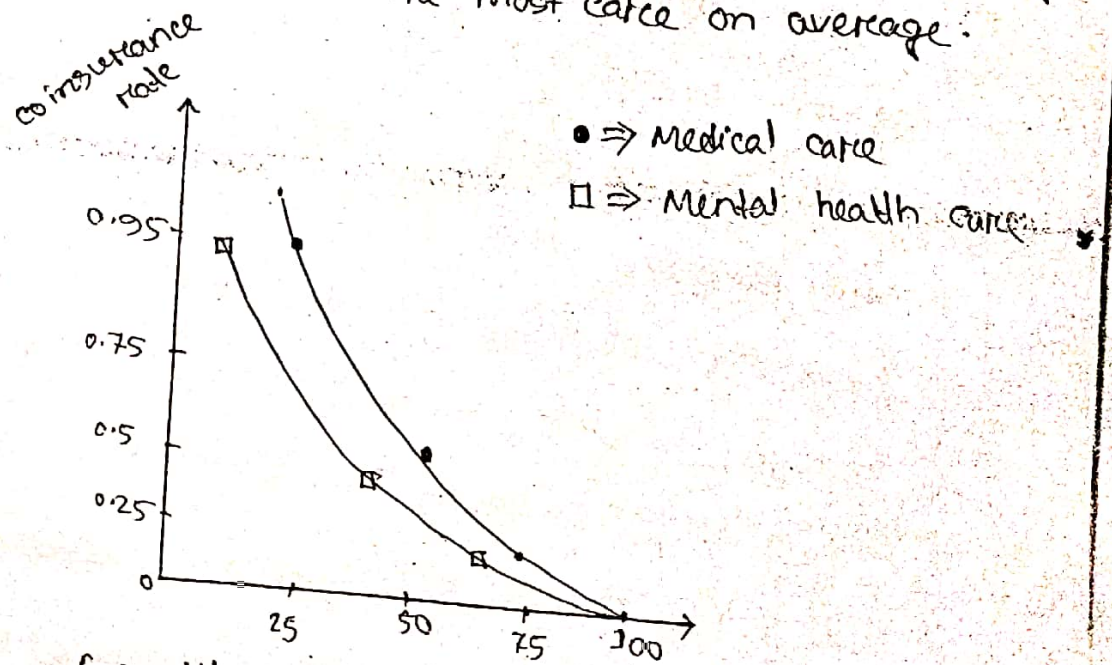


Fig: Demand response of ambulatory mental and medical care

This group is the group with free care. The vertical axis measures the economic incentives as indicated by the coinsurance rate - the percentage of the bill paid out directly by the consumer. Thus a higher coinsurance rate reflects a higher price to the consumer. The above figure shows people consuming more care as the care becomes less costly to the consumer in terms of dollars paid out of pocket. More importantly, the curve demonstrates that economic incentives do matter. Those facing higher prices demand less care.

Is health care different?

Although economics certainly applies to health care, it is more challenging to answer the question of how directly and simply it applies. Health care is different for the following reasons:

(i) Presence and extent of uncertainty:

When Nobel laureate Kenneth Arrow (1963) directed his attention to the economics of health, he helped establish health economics as a field. He stressed the prevalence of uncertainty in health care, on both the demand side and the supply side. Consumers are uncertain of their health status and need for health care in any coming period. This means that the demand for health care is irregular in nature from the individual's perspective; likewise the demand facing a health care firm is irregular.

Uncertainty is also prevalent on the supply side. Standard economic analysis often assumes that products and the pleasures that they bring, are well understood

by the purchasers. The purchase of steak, milk, new clothes etc provides expected well being that is easily known and understood. In contrast, several cases of product uncertainty exist in the health field. Consumers often do not know the expected outcomes of various treatments without physician's advice and in many cases physicians themselves can not predict the outcomes of treatments with certainty.

(ii) Prominence of insurance :

Consumers purchase insurance to guard against this uncertainty and risk. Because of health insurance, neither most Americans nor citizens of other countries pay directly for the full costs of their health care.

Suppose, in a specific year almost 52 percent of all personal health care expenditures were paid out of pocket meaning that 48 percent was paid by third party payers. Insurance changes the demand for care and it

potentially also changes the incentives of providers.

(iii) Problems of information:

Uncertainty can in part be attributed to lack of information. Sometimes information is unavailable to all parties concerned. For example, neither gynecologists ~~information is at~~ nor their patients may recognize the early stages of cervical cancer without a Pap smear. At other times, the information in question is known to some parties, but not to all, and then it is the asymmetry of information that is problematic.

The problems of information mean that careful economic analysis must modify their methods. They base their decisions on the characteristics of the goods, their prices and their abilities to bring pleasure to the buyers.

Consumers may not know which physicians or hospitals are good, capable or even competent. They may not know whether they are ill or what should be done if they are. This lack of information often makes a consumer dependent on the provider. The provider offers both the

information and the service leading to the possibility of conflicting interests.

(iv) Large role of non profit firms:

Economists often assume that firms maximize profits. The economist must analyze the establishment and perpetuation of non profit institutions and understand the differences in their behaviours from for profit firms. The citizenry is aware of the hospitals as centers of health care - the larger medical community views them as research centers.

(v) Restrictions and competitions:

The health sector has developed many practices that effectively restrict competition. These practices include licensure requirements for providers, restrictions on provider advertising and standards of ethical behaviour that enjoin providers from competing with each other. Regulation to promote quality or to curb costs also reduces the freedom of choice of providers and may influence competition.

(vi) Role of equity and need:

Poor health of another human being often evokes a feeling of concern that distinguishes health care from many other goods and services. Many advocates express ~~that~~ this feeling by saying that people ought to get the health care they need regardless of whether they can afford it.

(vii) Govt. subsidies and public provision:

In most countries, the govt. plays a major role in the provision of health services.

DEMAND FOR HEALTH CAPITAL

Explain the demand for health and health care according to Grossman and comparison between traditional demand and health care demand.

Grossman used the theory of human capital to explain the demand for health and health care. According to human capital theory, individuals invest in themselves through education, training and health to increase their earnings. Grossman shows the way in which many important aspects of health demand differ from the traditional approach to demand:

1. It is not medical care ~~as such~~ that the consumer wants, but rather health. People want health; they demand medical care inputs to produce it.

2. The consumer does not merely purchase health passively from the market. Instead, the consumer produces health, combining time devoted to health improving efforts with purchased medical inputs.

3. Health lasts for more than one period. It does not depreciate instantly and can be analyzed like a capital good.

4. Perhaps, most importantly, health can be treated both as a consumption good and an investment good. As a consumption good, health is desired because it makes

people feel better. As an investment good, health is desired because it increases the number of healthy days available to work and to earn income.

Explain the concepts of health capital

Following figure provides a simple ~~schematic~~ diagram that explains the concept of health capital. Consumers apply sets of health inputs, which might include market inputs of health care but also diet, exercise and time to their physical make-up thus making investments in health capital. These investments maintain or improve the consumers' stocks of health which in turn provide them ^{with} healthy days. Over time, the health stock may grow, remain constant or decline, either slowly with age or more with illness or injury.

From the figure we see how the end goal, "healthy days" guides consumer decisions as to how much time and money to invest in health stock. We will see that

the prices of health care, the people's wage rates, and their productivities in producing health, will dictate how resources are to be allocated among health capital and other goods and services that people buy.

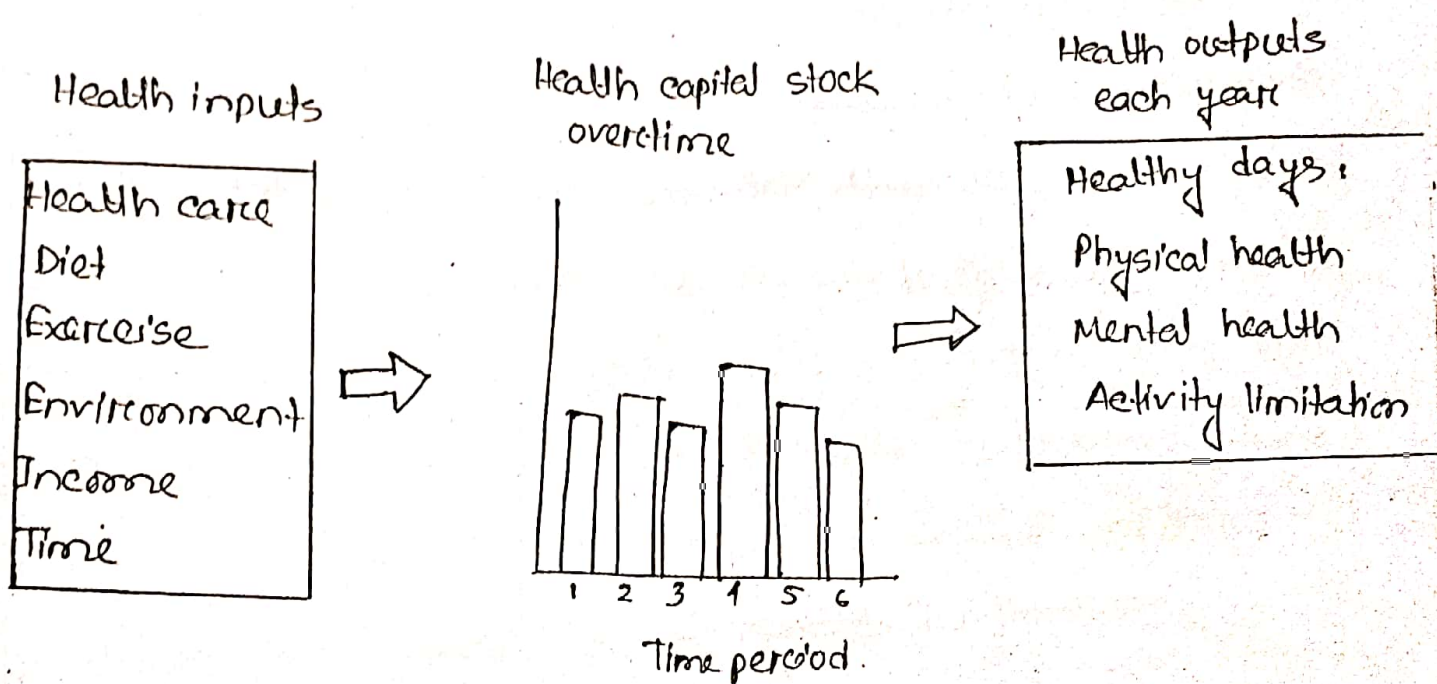


Fig: Investing in health capital

Consider a consumer, Edward, who buys market inputs (e.g. health care, food, clothing) and combines them with his own time to produce the stock of health capital that produces services that increase his utility. Edward uses market inputs and personal time both to invest in his

stock of health and to produce other things that he likes.

These other items include virtually all other things that Edward does. They include time spent watching television, reading, playing with and teaching his children, preparing meals, baking bread, a composite of other things people do with leisure time. We may call this composite home good.

Explain the health investment function and home good function in the light of Grossman model.

An increment to capital stock, such as health, is called an investment. Consider a consumer Edward, who buys market inputs (e.g. medical care, food, clothing) and combines them with his own time to produce the stock of health capital that produces services that increase his utility. Edward uses market inputs and personal time both to invest in his stock of health ^{and} to produce other things that he likes. During each period, Edward produces an investment in health, I . Thus,

Health investment, $I = I(M, T_H)$

Here, M = Market health inputs (providers' service, drugs)

T_H = Time spent improving health.

This function indicates that, increased amount of M and T_H increase I .

There are also other items include virtually all other things that Edward does. They include time spent

watching television, reading, playing with and teaching his children, preparing meals, baking bread or watching the sun set, a composite of other things people do with leisure time. we shall call this composite home good B .

Home good function, $B = B(x, T_B)$

Here, x = market purchased goods.

T_B = Time for producing home good.

Home good (B) is produced with time, T_B and market purchased goods, x . This function indicate that increased amounts of x and T_B increase B .

Thus, Edward is using money to buy health care inputs, M or home good inputs, x . He is using leisure time either for health care (T_H) or for producing the home good (T_B).

In this model, Edward's ultimate resource is his own time. Assume that, Edward has 365 days available in the year. To buy market goods such as medical care, M , or other goods x , he must trade some of his time for income,

that is he must work at a job. Call his time devoted to work T_w , we realize that some of his time during each year will be taken over by ill health or T_L . Thus, we account for his total time in the following manner.

$$\begin{aligned} \text{Total time} = T = 365 \text{ days} &= T_H (\text{Improving health}) + \\ &T_B (\text{Producing home goods}) + \\ &T_L (\text{lost to illness}) + T_w (\text{working}). \end{aligned}$$

1st tutorial

How would you explain the potential use of a person's time by the labor-leisure trade off?

The potential uses of a person's time are illustrated by the labor-leisure trade off. In the following figure, the x-axis represents Ed's work and leisure time. Suppose that a person considers his time spent creating health investment to be "health-improvement time" and that he calls T_H his leisure time. Assume further that the number of days lost to ill health and the number of days spent on health enhancing activity are fixed.

variable, $\left\{ \begin{array}{l} T_{Lo} = \text{Time lost} \\ T_{Ho} = \text{Time spent on healthy activities} \end{array} \right.$

The maximum amount of time that he has available to use either for work, T_w or leisure, T_B 's,

$$\begin{aligned} \text{Time available for work or leisure} &= 365 - T_{Ho} - T_{Lo} \\ &= T_w + T_B \end{aligned}$$

Leisure time T_B is measured toward the right while time spent at work, T_w is measured toward the

left. If the person chooses leisure time OA , then he has simultaneously chosen the amount of time of work indicated by AS . These are indicated in the following figure

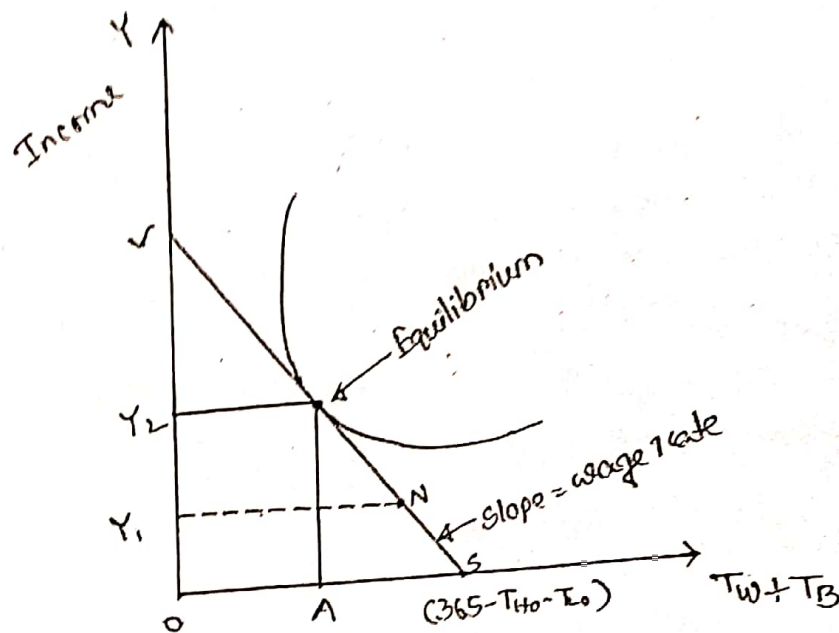


Fig: Labor-leisure trade off

The person's total amount of time available for either work or leisure is given by point S . If he choose point S for the period, he would choose to spend all his available time in leisure. The Y -axis represents Income, which can be obtained through work. This Income can be used to purchase either market health goods or other market health goods. Thus if he chooses point S , he will not be able to

purchase market goods because he has no wage.

If the person gives up one day of leisure by spending that day at work, to point N, he will generate income equal to OY_1 , which represents his daily wage. In economic terms, this quantity represents income divided by days worked; that is the daily wage = $\frac{\text{Income}}{\text{Days worked}}$.

The slope of the line vs depicting the labor-leisure trade off reflects the wage rate. In equilibrium, his trade off of leisure and income is the same as the market trade off which is the wage rate. Hence, he takes OA amount of leisure and trade AS amount of leisure for income OY_2 .

Explain the impact of an increase in activities to improve health on income and leisure.

An increase in activities to improve health, increased the time spent investing in health status. Time spent on health producing activities T_H is increased from T_{H0} to T_{H1} . Correspondingly suppose that the number of days lost to ill health has been reduced from T_{L0} to T_{L1} . Time spent producing health reduces time available for other activities. On the other hand time spent on health investment increases health stock and in turn reduces the time lost to illness.

If the net effect is a gain in available time, then this illustrates the pure investment aspect of health demand. The health investments "pay off" in terms that not only add to potential leisure but that also increase the potential time income, shifting the income-leisure line outward from VS to RQ . The expenditure of time for health producing activities may later improve available hours of productive activity.

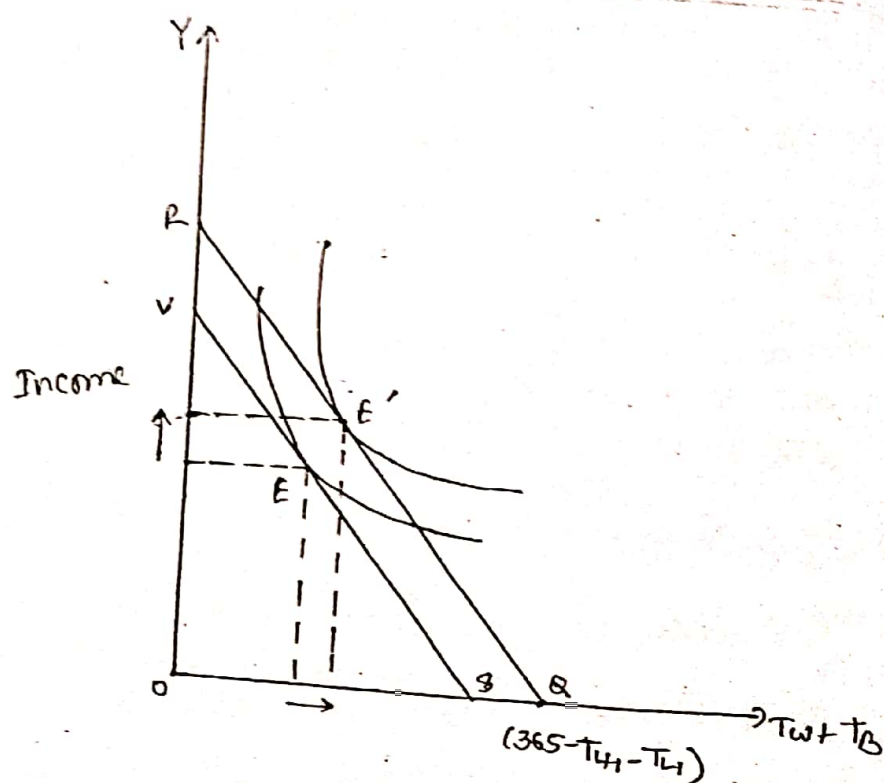


Fig: Increased amount of healthy time due to investment

As a result of investment, utility increases moving from point E to point E' . Not only does investment in health lead to ~~be feeling~~ better, but also leads to more future income and may lead to more leisure as well.

The improvement in health status also might increase productivity at work, perhaps resulting in a higher wage.

Production of healthy days.

Health is a productive good that produces a single output, healthy days. This production function relationship is illustrated in the following figure. The horizontal axis measures health stock in a given period. The greater one's stock of health, the greater the number of healthy days, up to a natural maximum of 365 days. The bowed shape of the curve illustrates the law of diminishing marginal returns. The concept of a health stock minimum shown as H_{min} . At this point, production of healthy days drops to zero indicating death.

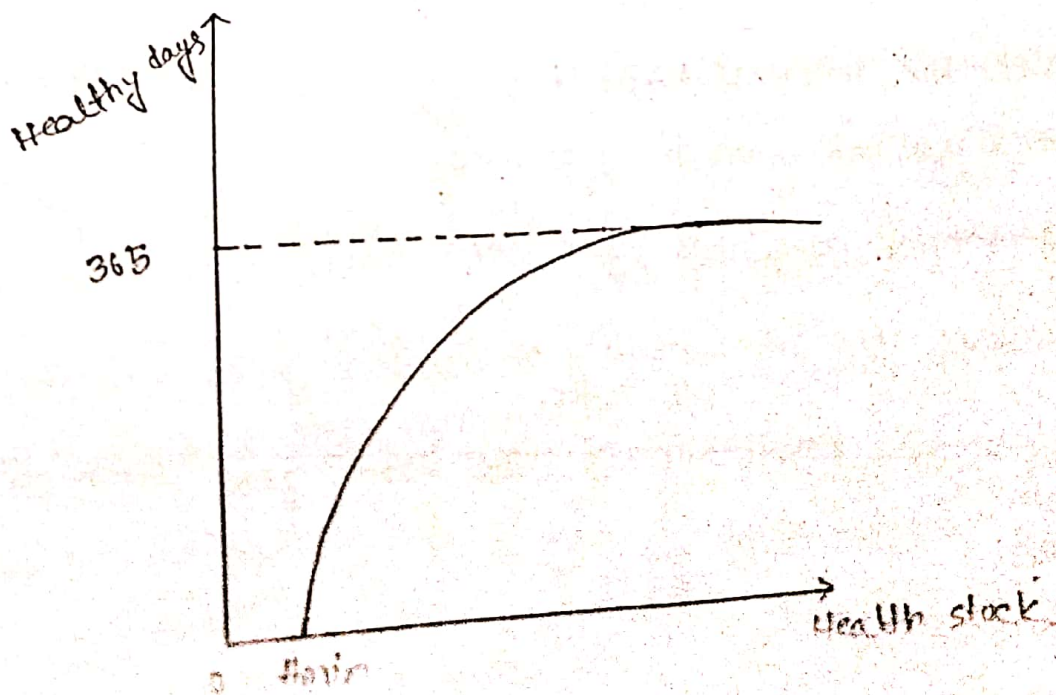


Fig: Production of healthy days.

Production of health and home goods / Allocation of resources between health and market good (such as bread)

Consider the possibilities for producing health, H , and home good B given the amount of time variable. Following figure shows the production possibilities trade-off. The curve differs from the usual production possibilities curve in several respects. First, from point A to point C , health improvements increase the amounts of the home good B and health H attainable. It is necessary to increase health beyond H_{min} in order to obtain income and leisure time from which to produce B .

Moving along the production possibilities curve, shifts the uses of available time and distributes the purchases of market goods. The move from E to C indicates that more available time for health. This movement also increases the availability of market goods and time used to increase production of the home good such as baking bread.

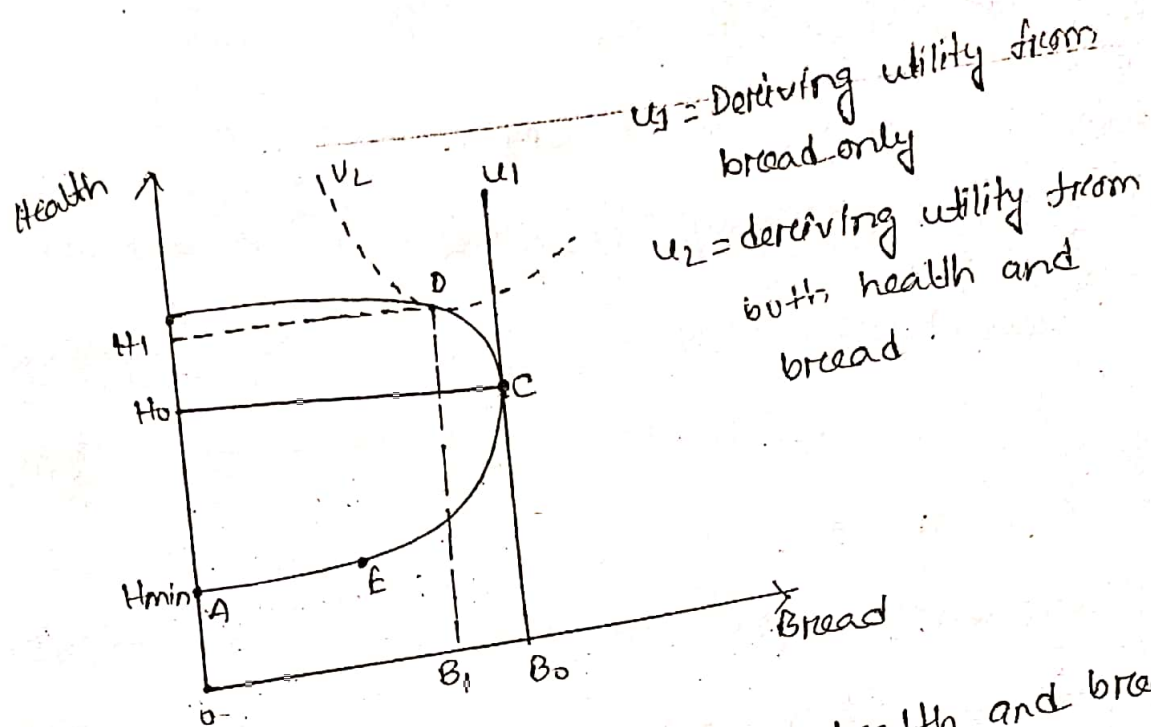


Fig: Allocation of resources between health and bread

Suppose that a person desires health solely for its effect on the ability to produce income and the leisure time to produce the home good bread, B . This would imply that his indifference curves between H and B would be vertical lines. In such a case he would maximize his utility by producing as much B as possible. The utility maximizing would be at point C , a point of tangency between indifference curve U_1 and the production possibilities curve. He would derive no utility from the home good and H_0 of his

Now assume instead that Ed achieves utility not only from producing B, but also directly from health itself. In this case his indifference curve U_2 is sloping downward from left to right. It is more realistic to say that he values health as a consumption good and as an investment in productive capacity. The former suggests that he enjoys feeling healthy; the latter that feeling healthy makes him more productive, thus allowing him to earn more.

Continuing ...



Chapter 2

Microeconomic tools for health economics

Microeconomic

Using supply and demand graph and assuming competitive markets, show and explain the effects on equilibrium price and quantity of the following:

- (a) Increased graduation of new doctors on the market for physician services.
- (b) Increased cost of medical education on the market for physician services.
- (c) The virtual elimination of smoking on the market for hospital services.
- (d) Decreased price of Napa on the market for Ace
(Hints: Napa and Ace are substitutes)
- (e) A technological improvement that reduces the cost of producing x-rays on the market for physician clinic services.

Solution:

- (a) Increased graduation of new doctors on the market for physician services causes increases the supply of physician services. So, this increases the quantity of visit and decreases the price of visit.

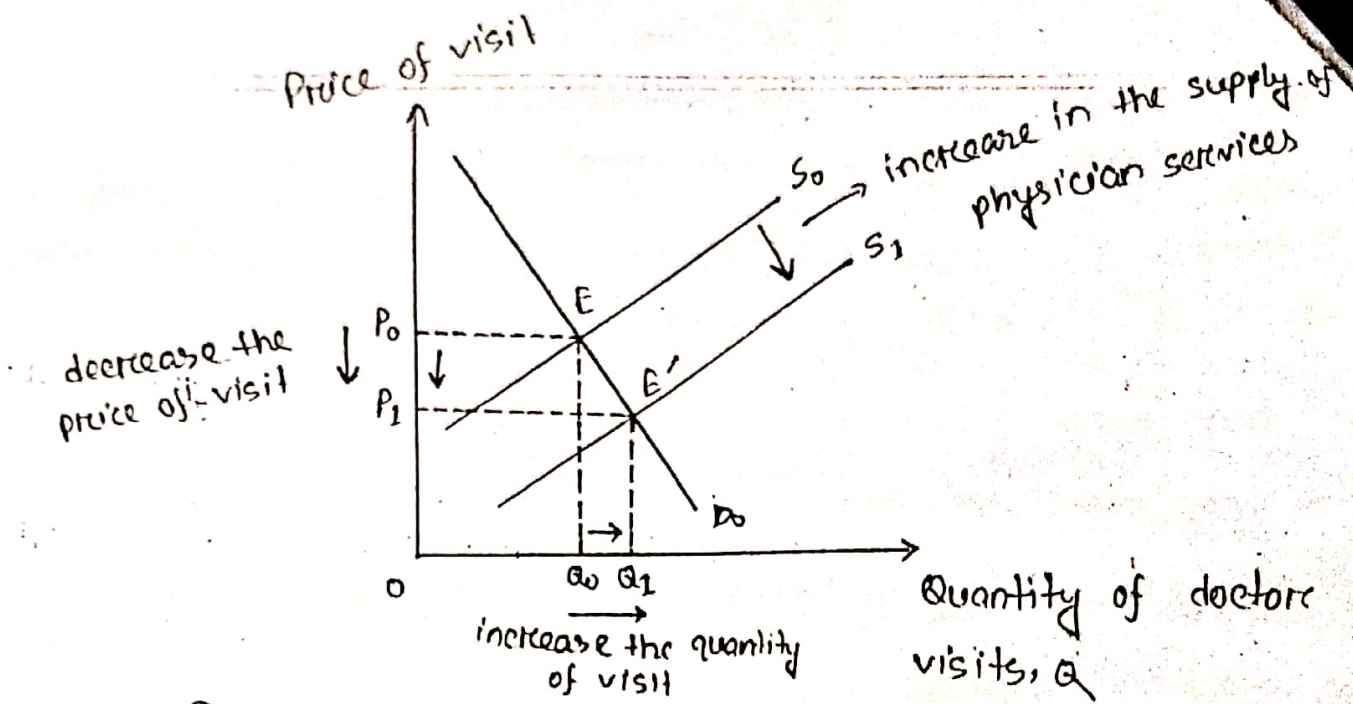


Fig: Market for physician services.

In the figure, when the graduation of new doctors increases, the supply of physician services increases. The supply curve for physician services shifts right from S_0 to S_1 . As a result, the quantity of doctors visits increases from Q_0 to Q_1 and this leads to decrease in the price of visit from P_0 to P_1 .

(b) If the cost of medical education increases, it will lead to decrease in the supply of physical services. Because of high cost of medical education, high price of medical services is charged

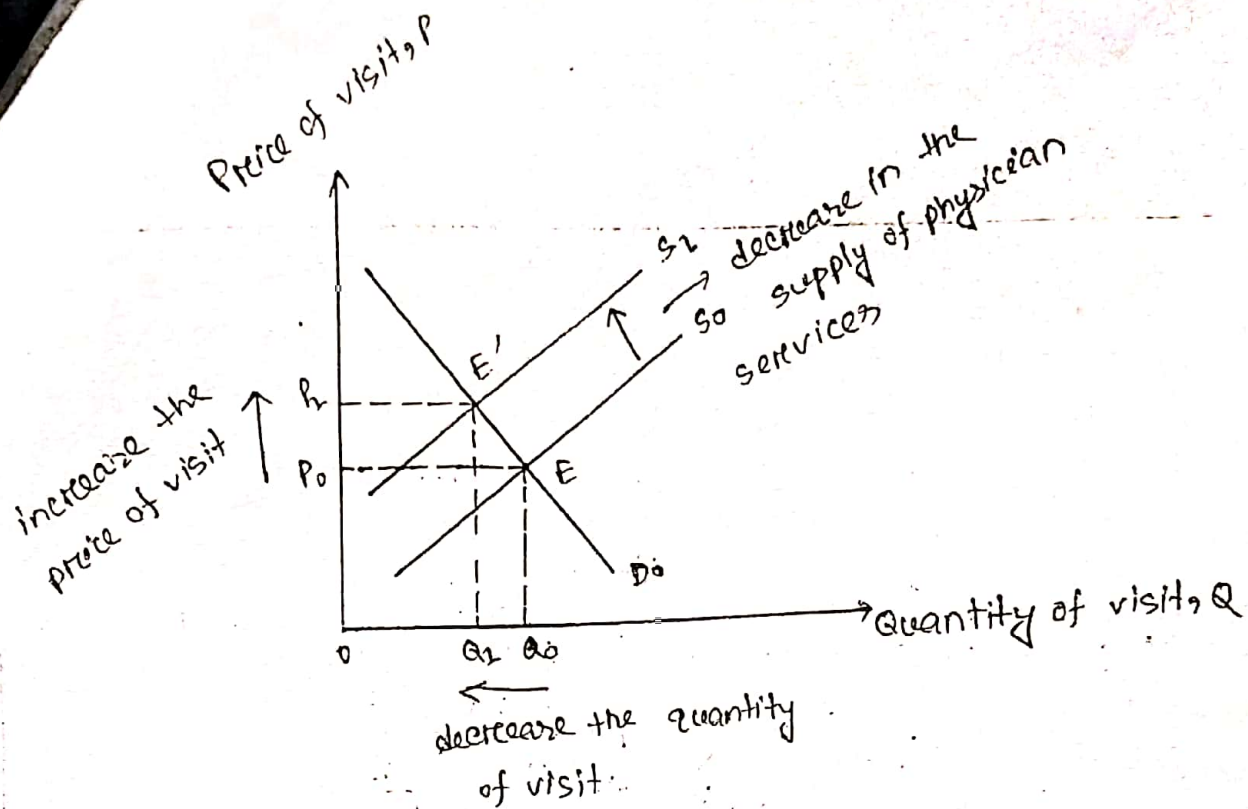


Fig: Market for physician services.

When the cost of medical education increases, the supply of physician service or medical service decreases. This shifts the supply curve to the left from S_0 to S_1 . As a result, initially equilibrium point $E(Q_0, P_0)$, where quantity of visits was Q_0 and price of visit was P_0 . After the decrease of the supply of physician services, there creates a new equilibrium point E' , where the quantity of visit decreases from Q_0 to Q_1 and price of visit increases from P_0 to P_1 .

© If people eliminate smoking, they will demand less hospital services. As their demand falls for hospital services, the price of per bed used reduces.

Here, suppose that hospital service equals to the number of beds or cabin used.

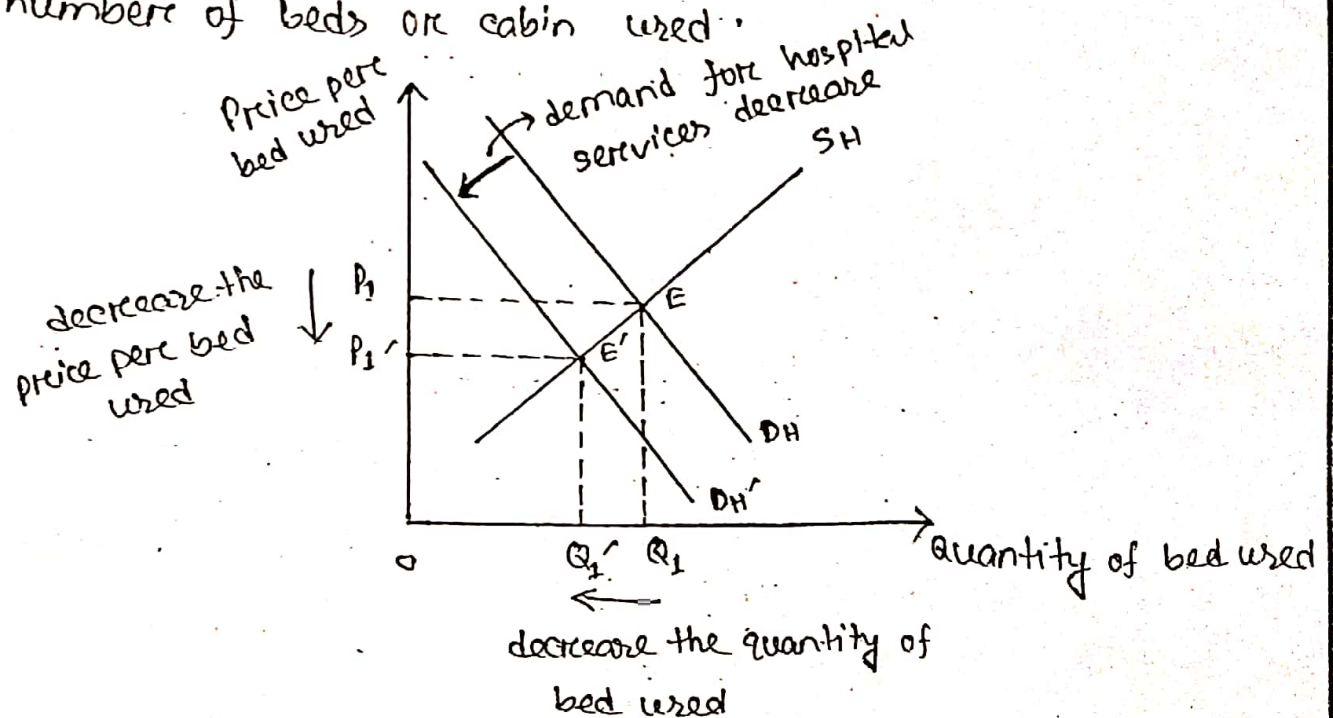


Fig: Market for hospital service.

As people reduce smoking, the demand for hospital services also reduce decrease. So the demand curve for hospital services shifts to the ~~right~~ ^{left} from DH to DH' . Initially, at DH demand curve, the ~~dema~~ quantity of bed used was

Q_1 and the price per bed used was P_1 . With the shift of the demand curve, the equilibrium point shifts to E' , where the quantity of bed used decrease to Q_1' and the price per bed used also decrease to P_1' .

(d) As the Napa and Ace are substitutes, the decrease in the price of Napa will lead to a decrease in the demand for Ace.

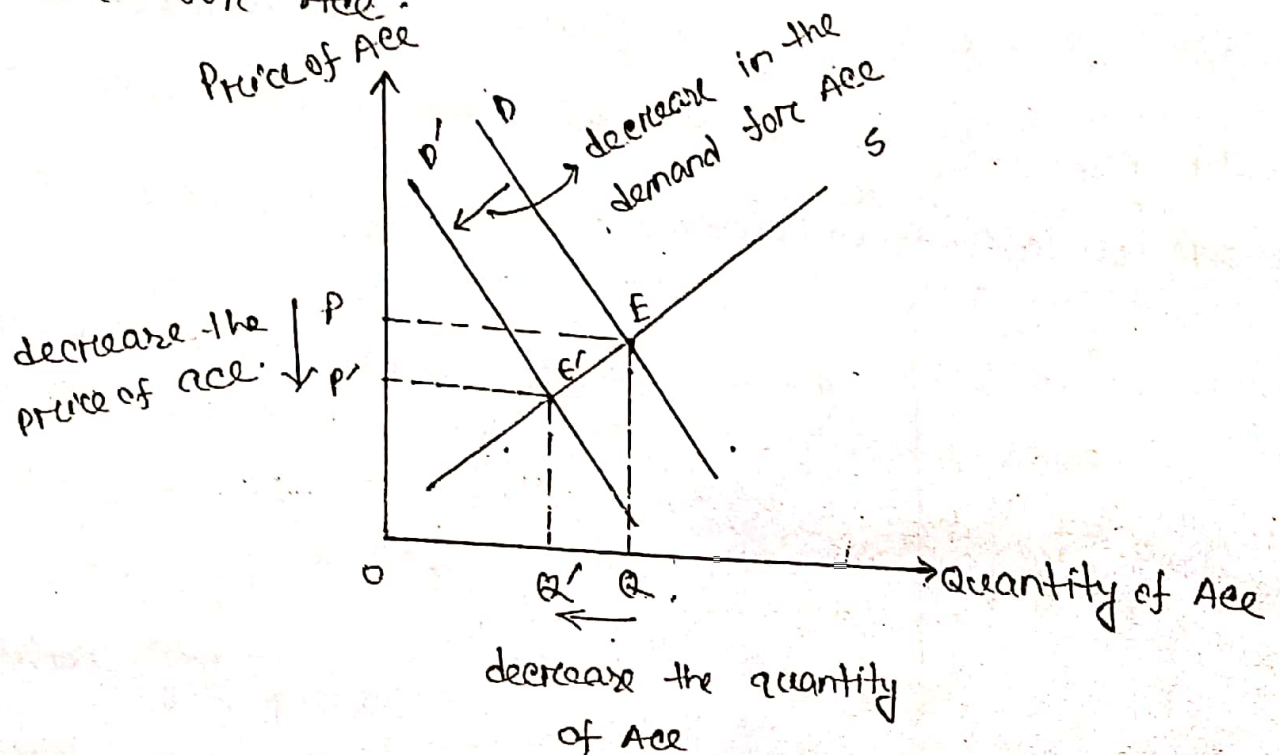


Fig: Market for Ace.

As the price of Napa decreases, the demand for Napa increases. On the other hand the demand for Ace decreases because Napa and Ace are substitutes.

Initially, the equilibrium was at point E, where the demand for the AEC was Q and price was P. With the decrease in the price of Napa, the demand curve for AEC shifts to the left from D to D'. Note the equilibrium point is E' where the quantity of AEC decrease to Q' and price decrease to P'.

② A technological improvement that reduces the cost of producing x-rays on the market for physician clinic services leads to an increase in the supply of medical services.

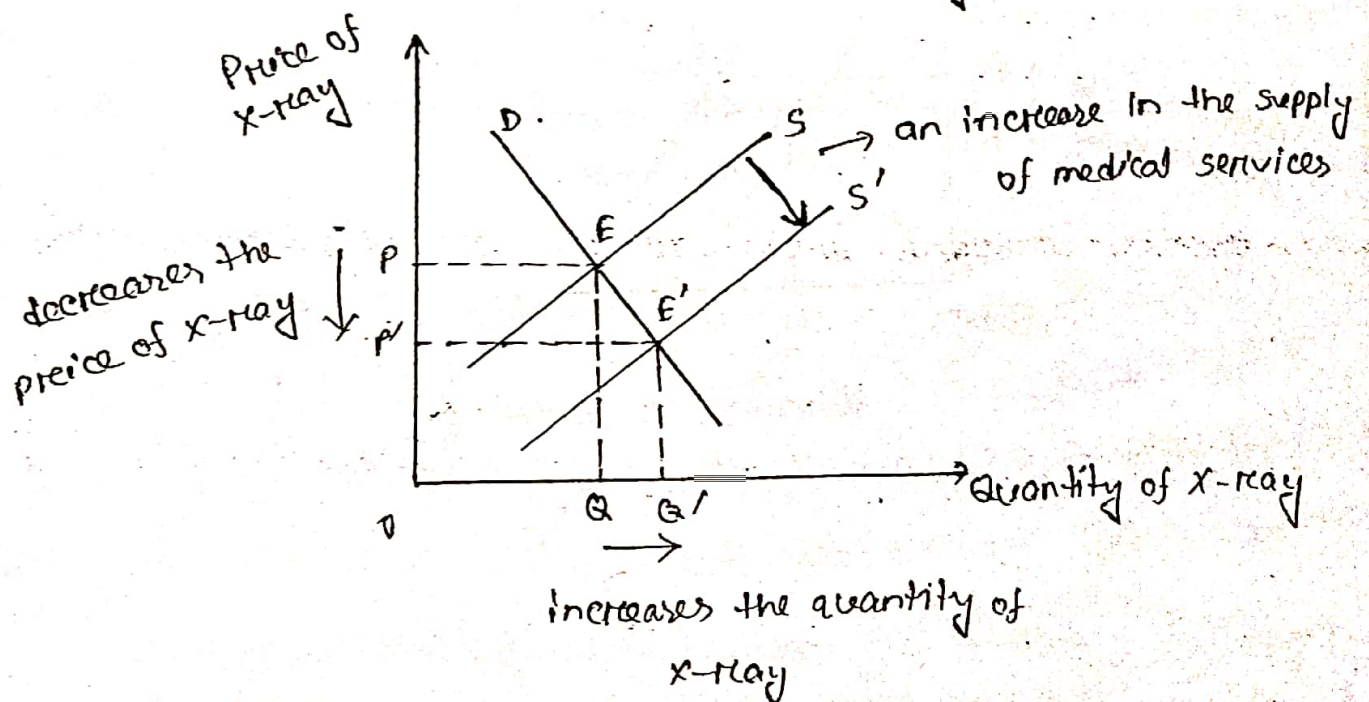


Fig: Medical service market

~~As the supply curve shifts from~~

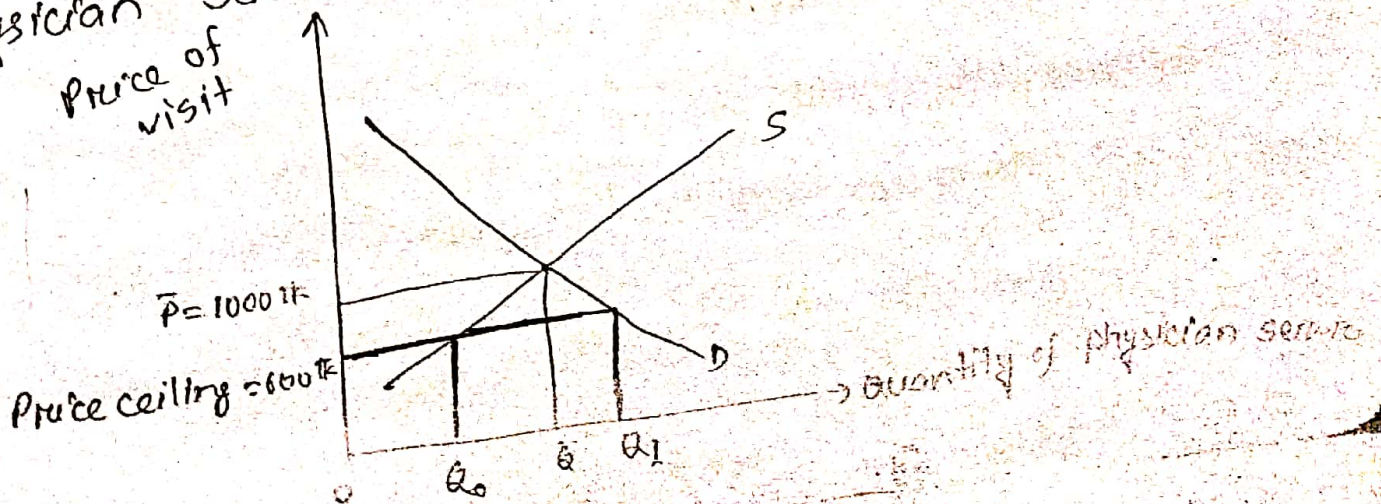
As the supply increases and the supply curve shifts from S to S' , the price of x-ray decreases from P to P' and the quantity of x-ray increases from Q to Q' .

The price ceiling placed on physician fees on the market for physician services.

Without price ceiling the quantity of visit was Q at the 1000 TK price of per visit. Suppose govt. placed a price ceiling on physician fees of 600 TK.

This creates an excess demand for physician service, Q_1 . But the supply of the physician services decreases to Q_0 .

This is the demand for physician services decreases. As the demand for physician services decreases and supply decreases, this decreases the quality of physician service.



Demand for Health capital

The cost of capital.

We demand health because it helps us earn more and feel better. Suppose that an x-ray machine costs \$100,000 and that its price does not change over time. Suppose that the annual income attributable to the use of the x-ray machine is \$20,000. Consider the alternative: Instead of purchasing the x-ray machine, the clinic ~~can~~ could have put the \$100,000 in a savings account, at 5 percent interest rate, yielding the following:

$$100,000 \times 1.05 = 105,000 \text{ at the end of Year 1}$$

$$105,000 \times 1.05 = 110,250 \text{ at the end of Year 2}$$

$$110,250 \times 1.05 = 115,763 \text{ at the end of Year 3}$$

$$115,763 \times 1.05 = 121,551 \text{ at the end of Year 4}$$

$$121,551 \times 1.05 = 127,629 \text{ at the end of Year 5}$$

For the investment in an x-ray machine to be desirable by these criteria, it should provide at least \$27,629 in incremental revenue over the five years.

Suppose that the clinic knows that the x-ray

machine will wear out, so that it will be worth only half its original value in five years. The clinic must earn enough not only to cover the opportunity cost from the bank, but also to maintain the value of the machine. For an investment in an x-ray machine to be worthwhile when it must not only earn the competitive 5 percent return per year but it also must provide enough return to cover depreciation. This suggests that the cost of holding this capital good for any one year as well as over time, will equal the opportunity cost of the capital (interest rate foregone) plus the depreciation.

Marginal efficiency of investment (MEI) and rate of return
optimal health stock.

The marginal efficiency of investment curve shows how responsive investment is to a change in interest rates. The MEI can be described in terms of the X-ray machine example. A busy clinic may wish to own more than one X-ray machine. The first machine purchased yield a return of \$20,000. The rate of return is $\frac{\$20,000}{\$100,000} = 20$ percent per year. The management would buy this machine if the incremental revenue brought in covered its opportunity cost of capital and the depreciation. In terms of rates, management would choose to own the first X-ray machine as long as the rate of return, 20 percent, were greater than the interest rate plus the depreciation rate.

If the management considered owning two machines, it would discover that the rate of return on second X-ray machine would probably be less than the first. This is

best understood by recognizing that a clinic buying only one x-ray machine would assign it to the highest priority uses, those with the highest rate of return. If the clinic were to add a second x-ray machine, then logically it could be assigned only to lesser-priority uses. Thus, the second machine would have a lower rate of return than the first. The clinic would then purchase the second machine only if its rate of return were still higher than interest plus depreciation.

The decreasing MEI:

In the following figure, the MEI curve, describes the pattern of rates of return; declining as the amount of investment increases. The cost of capital (interest rate, r + the depreciation rate, δ) is shown as the ~~lower~~ horizontal line labeled $(r + \delta)$. The optimum amount of capital demanded is thus H_0 , which represents the amount of capital at which the marginal efficiency of investment just equals the cost of

capital. This equilibrium occurs at point A.

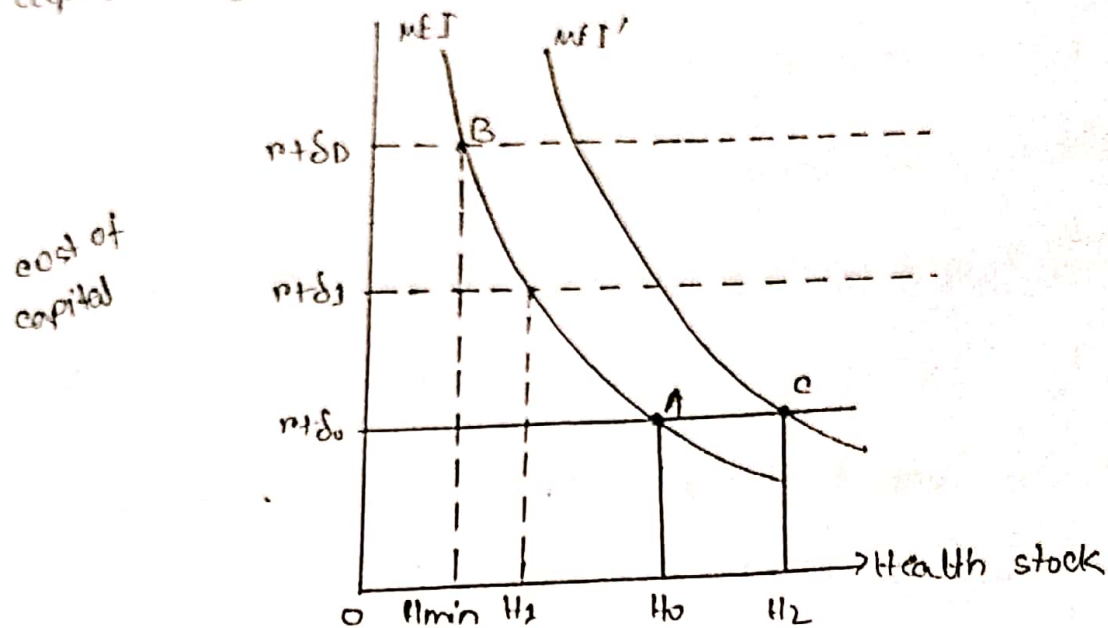


Fig: Optimal health stock.

The MEI curve for investments in health also would be downward sloping. This occurs because the production function for healthy days exhibits diminishing marginal returns. The cost of capital for health would similarly reflect the interest rate plus the rate of depreciation in health. Understand that a person's health like any capital good, also will depreciate over time. Thus, the optimal demand for health is likewise given at the intersection of the MEI curve and the cost of capital curve ($r + \delta_0$).

changes in equilibrium: Age, wage, education and uncertainty.

Age: Age is a key factor for changing health condition. Grossman indicates Health stock may depreciate faster during some periods of life and more slowly during others. Eventually as human ages, the depreciation rate is likely to increase. In other words, the health of older people is likely to deteriorate faster than the health of younger people.

Consider the effect of aging on optimal health stock. By hypothesis the depreciation rate δ increases with age from δ_0 to δ_1 and ultimately to δ_2 . This implies that the optimal health stock decreases with age. Optimal health at the younger age, H_0 is greater than H_1 , the optimal stock at the older age. Higher depreciation rates increase the cost of holding health capital stock. In old ages health depreciation rates are extremely high, δ_0 and optimal health stock falls to H_{min} at point B.

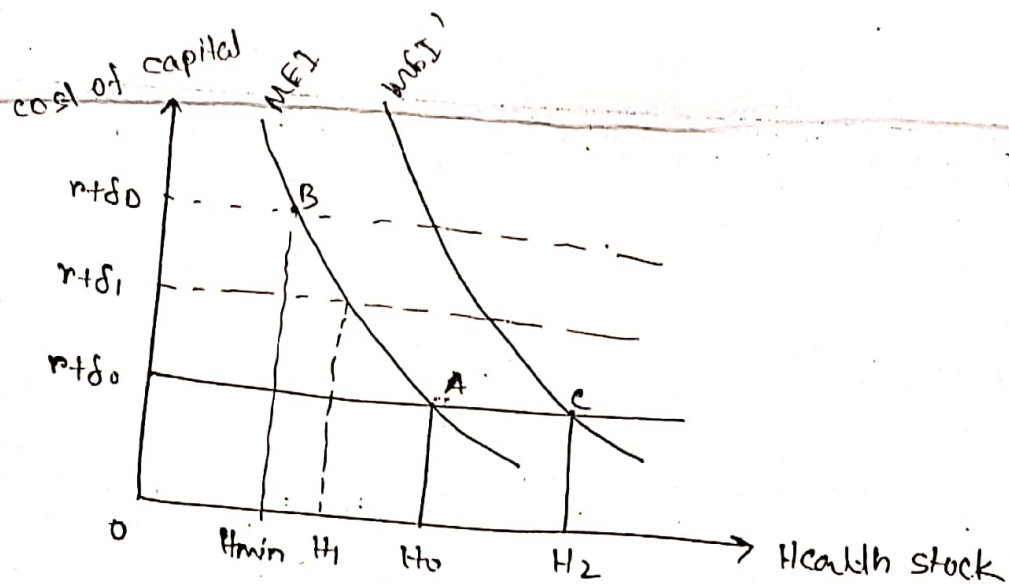


Fig: Optimal health stock

As the expected length of life decreases, the MEI curve shifts to the left, because the returns from an investment will last for a shorter period of time. Other features of the model suggest that people will increase their gross investment in health as they age. This suggests, in turn, that the elderly would demand more medical care than the young.

Wage rate:

The above figure illustrates the effect of a change in the wage rate on optimal level of investment. Increased wage rates increase the returns obtained

from healthy days. Thus higher wages imply a higher MEI curve or MEI' .

The original MEI curve describes the lower wage case and yields optimal stock health stock, H_0 . MEI' above MEI shows the marginal efficiency of investment for someone with higher wages. At new equilibrium point C, the higher wage will imply a higher optimal level of health stock, H_2 . When person retires, his wage effectively drops to zero.

Education

Education is of special interest in the demand for health. Higher education is most often related to better health and most economists believe that education may improve the efficiency with which one can produce investments to health and the home good.

The effect of education is illustrated in the figure. Here the MEI curve illustrates the marginal efficiency

investment for the consumer with a low level of education. while the MEI' curve illustrates the same person with a higher level of education. This model indicates that because education raises the marginal product of the direct inputs, it reduces the quantity of these inputs required to produce a given amount of gross investment.

It follows that given investments in health can be generated at less cost for educated people and thus they experience higher rates of return to a given stock of health. The health, as shown is that the more educated people will choose higher optimal health stocks, H_2 than the less educated who will choose H_0 . Educated people tend to be significantly healthier.

Educated people are also likely to recognize the benefits of improved health. they may enjoy eating nutritious food or doing physical exercise. They may recognize the dangers of smoking, and the long term.